Nuclear Magic Numbers from Torus Topology*

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Abstract

We derive all seven nuclear magic numbers $(2, 8, 20, 28, 50, 82, 126)$ from the torus topology of the Null Worldtube (NWT) model, with zero free parameters beyond those established in previous work [1, 2]. The derivation chain is: (i) the pion mass follows from the torus energy scale, $m_\pi = 2m_e/\alpha = 140.1$ MeV; (ii) the nuclear potential depth follows from one-pion exchange on the $k = 3$ torus, $V_0 = 50.2$ MeV; (iii) the torus metric decomposes the Dirac mean field into scalar and vector components, giving $V - S = (2k^2 - 1)V_0 = 853$ MeV and an effective nucleon mass $M^* = M_N - k^2 V_0 = 487$ MeV; (iv) the resulting spin-orbit potential, solved in a Woods–Saxon shell model, reproduces all seven experimental shell closures. As a non-trivial test, we trace the four natural radioactive decay series (^{232}Th, ^{237}Np, ^{238}U, ^{235}U) using NWT-derived nuclear masses with shell corrections, obtaining the correct stable endpoints (Pb/Bi isotopes) in all four cases. The doubly-magic nucleus ^{208}Pb — the terminus of three of the four series — owes its exceptional stability to shell closures at $Z = 82$ and $N = 126$, which are direct consequences of the $k = 3$ torus geometry. This extends the NWT program from leptons and mesons into nuclear structure, demonstrating that a single topological framework can account for phenomena spanning twelve orders of magnitude in energy scale.

1 Introduction

The nuclear magic numbers — $2, 8, 20, 28, 50, 82, 126$ — are among the oldest empirical facts in nuclear physics. Nuclei with a magic number of protons or neutrons exhibit enhanced binding energy, higher first-excited-state energies, and anomalously large abundances in nucleosynthesis. The doubly-magic nuclei (^{4}He, ^{16}O, ^{40}Ca, ^{48}Ca, ^{208}Pb) are exceptionally stable, and ^{208}Pb in particular serves as the endpoint of three of the four natural radioactive decay series [3, 4].

The explanation of magic numbers was one of the great triumphs of twentieth-century nuclear physics. Mayer [3] and, independently, Haxel, Jensen, and Suess [4] showed that a strong spin-orbit interaction, added to the nuclear shell model, splits orbital degeneracies in exactly the pattern needed to produce the observed shell closures. However, the spin-orbit force itself was introduced phenomenologically — its strength was adjusted to reproduce the data, not derived from a deeper principle.

A microscopic understanding came decades later from the relativistic mean-field (Walecka) model [5, 6]. In this framework, nucleons exchange scalar (σ) and vector (ω) mesons, producing large, nearly cancelling potentials: $S \approx -400$ MeV (scalar attraction) and $V \approx +350$ MeV (vector repulsion). The binding energy depends on $V + S \approx -50$ MeV, while the spin-orbit force depends on $V - S \approx 750$ MeV. The near-cancellation in binding with a large sum in spin-orbit is the essential physics. But the Walecka model takes the meson masses and couplings as inputs — it does not explain *why* the scalar-vector splitting has the values it does.

In this paper we show that the Null Worldtube (NWT) model provides exactly this explanation. In the NWT framework [1, 2], elementary particles are photons confined to torus knot trajectories.

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The torus aspect ratio $k = R/r$ and toroidal winding number p determine all particle properties. The nuclear force arises from one-pion exchange, where the pion itself is a torus excitation with mass $m_\pi = 2m_e/\alpha$. We show that the torus metric naturally decomposes the Dirac mean field into scalar and vector components with a fixed ratio determined by k, yielding the spin-orbit force with no free parameters.

The structure of the paper is as follows. Section 2 derives the pion mass and nuclear potential depth from the NWT energy scale. Section 3 derives the scalar-vector decomposition from the torus metric. Section 4 constructs the spin-orbit potential and solves the Woods–Saxon shell model. Section 5 presents the resulting magic numbers and compares with experiment. Section 6 traces the four natural decay chains as a non-trivial test. Section 7 discusses the implications for the NWT program and the connection between electromagnetic and nuclear scales.

2 From the Torus to the Nuclear Potential

2.1 The pion mass

In the NWT framework, the fundamental energy scale associated with the torus is the ratio of the electron mass to the fine-structure constant:

$$\Lambda_\pi \equiv \frac{m_e}{\alpha} = \frac{0.51100 \text{ MeV}}{1/137.036} = 70.03 \text{ MeV}. \tag{1}$$

This scale arises because the electron's Compton wavelength $\lambda\!\!\!^{-}_e = \hbar/m_e c$ is related to the torus major radius by $R_e = \lambda\!\!\!^{-}_e/p$, and the electromagnetic coupling on the torus introduces a factor of α that connects the confined photon energy to the characteristic nuclear scale.

The pion, as the lightest meson mediating the nuclear force, corresponds to the lowest-lying excitation of the torus field. Its mass is:

$$m_\pi = 2\Lambda_\pi = \frac{2m_e}{\alpha} = 140.06 \text{ MeV}, \tag{2}$$

compared to the experimental value $m_{\pi^\pm} = 139.57$ MeV (0.35% error). The factor of 2 arises from the $(2, 1)$ winding of the nucleon: the pion excitation spans one full toroidal circuit of a $p = 2$ configuration.

2.2 The pion decay constant

The pion decay constant is determined by the torus geometry through:

$$f_\pi = \frac{k+1}{k}\,\Lambda_\pi = \frac{4}{3} \times 70.03 = 93.37 \text{ MeV}, \tag{3}$$

compared to the experimental value $f_\pi = 92.07$ MeV (1.4% error). The factor $(k+1)/k$ is the ratio of the torus circumference at the outer equator to the mean circumference, reflecting the asymmetry of the pion wavefunction on the curved torus surface.

2.3 The axial coupling and pion-nucleon vertex

The torus curvature contributes to the axial-vector coupling constant through:

$$g_A = 1 + \frac{\kappa}{2p}, \qquad \kappa \equiv \frac{k}{\sqrt{k^2-1}} = \frac{3}{2\sqrt{2}} = 1.061, \tag{4}$$

giving $g_A = 1 + 1.061/4 = 1.265$, compared to the experimental value $g_A = 1.2754$ (0.8% error). From the Goldberger–Treiman relation:

$$g_{\pi NN} = \frac{g_A\, M_N}{f_\pi} = \frac{1.265 \times 938.9}{93.37} = 12.72, \tag{5}$$

compared to the empirical value $g_{\pi NN} \approx 13.1$ (2.9% error).

2.4 The nuclear potential depth

The depth of the central nuclear potential follows from one-pion exchange in the Born approximation:

$$V_0 = \frac{g_{\pi NN}^2}{4\pi} \left(\frac{m_\pi}{2M_N}\right)^2 m_\pi \, C_{\text{eff}}, \tag{6}$$

where $C_{\text{eff}} = 2p + 1 = 5$ is an effective combinatorial factor counting the number of independent pion-exchange channels on the $(2, 1)$ torus knot. Evaluating:

$$f^2/(4\pi) = \frac{g_{\pi NN}^2}{4\pi} \left(\frac{m_\pi}{2M_N}\right)^2 = \frac{12.72^2}{4\pi} \times \left(\frac{140.1}{1877.8}\right)^2 = 0.0716, \tag{7}$$

and therefore:

$$V_0 = 0.0716 \times 140.1 \times 5 = 50.2 \text{ MeV}. \tag{8}$$

This is the depth of the Woods–Saxon potential used in all subsequent calculations. The standard empirical value is $V_0 \approx 50$–55 MeV.

3 Scalar-Vector Decomposition from the Torus Metric

3.1 The torus metric and Dirac decomposition

The toroidal metric component $g_{\phi\phi}(\theta) = (R + r\cos\theta)^2$ can be written as $R^2(1 + \cos\theta/k)^2$. When the Dirac equation is formulated on this background, the mean-field potential naturally separates into scalar (mass-like) and vector (energy-like) components.

The scalar component couples to the spatial metric volume element, which averages over the poloidal angle to give a contribution proportional to k^2:

$$S = -k^2 \, V_0 = -9 \times 50.2 = -452 \text{ MeV}. \tag{9}$$

The vector component couples to the propagation distance, determined by the difference $R^2 - r^2 = R^2(1 - 1/k^2)$, proportional to $k^2 - 1$:

$$V = (k^2 - 1) \, V_0 = 8 \times 50.2 = +401 \text{ MeV}. \tag{10}$$

3.2 Binding and spin-orbit: the key identity

The binding potential is the sum:

$$V + S = \left[(k^2 - 1) - k^2\right] V_0 = -V_0 = -50.2 \text{ MeV}, \tag{11}$$

which is the large cancellation that gives a shallow nuclear potential — a necessary feature of nuclear saturation.

The spin-orbit relevant combination is the *difference*:

$$V - S = \left[(k^2 - 1) + k^2\right] V_0 = (2k^2 - 1) \, V_0 = 17 \times 50.2 = 853 \text{ MeV}. \tag{12}$$

The effective nucleon mass in the medium is:

$$M^* = M_N + S = M_N - k^2 V_0 = 939 - 452 = 487 \text{ MeV}, \tag{13}$$

corresponding to $M^*/M_N = 0.52$.

The key result is that the *ratio* $(V - S)/V_0 = 2k^2 - 1 = 17$ is fixed entirely by the torus aspect ratio $k = 3$. In the Walecka model, this ratio is $750/50 = 15$ — the NWT prediction is 13% higher, but the parametric structure is identical. The large spin-orbit force is a direct geometric consequence of the torus curvature.

Table 1: Scalar-vector decomposition: NWT ($k = 3$) vs. Walecka model.

Quantity	NWT ($k = 3$)	Walecka	Error
S (MeV)	-452	-400	13%
V (MeV)	$+401$	$+350$	15%
$V + S$ (MeV)	-50	-50	0%
$V - S$ (MeV)	853	750	14%
M^* (MeV)	487	539	10%
M^*/M_N	0.52	0.57	10%

3.3 Enhancement over Thomas precession

The Thomas spin-orbit coupling alone gives $V_{\rm SO} \propto V_0/(2M_N^2)$, which is too weak by a factor of ~ 40 to explain the magic numbers. The enhancement factor in the NWT framework is:

$$\eta = \frac{V - S}{V_0}\left(\frac{M_N}{M^*}\right)^2 = 17 \times \left(\frac{939}{487}\right)^2 = 63, \tag{14}$$

compared to $\sim$ 40–60 from phenomenological fits. The geometric origin is transparent: the factor $2k^2 - 1$ comes from the torus metric, and the $(M_N/M^*)^2$ enhancement comes from the reduced effective mass.

4 The Spin-Orbit Potential and Shell Model

4.1 Spin-orbit potential

The nonrelativistic reduction of the Dirac equation in scalar and vector mean fields gives the spin-orbit Hamiltonian [6]:

$$H_{\rm SO} = \frac{V - S}{4M^{*2}c^2}\,\frac{1}{r}\,\frac{dU}{dr}\,\boldsymbol{\ell}\cdot\mathbf{s}, \tag{15}$$

where $U(r)$ is the Woods–Saxon central potential with depth V_0, radius parameter $r_0 = 1.27$ fm, and diffuseness $a = 0.67$ fm:

$$U(r) = \frac{-V_0}{1 + \exp[(r - r_0 A^{1/3})/a]}. \tag{16}$$

The spin-orbit strength in MeV·fm^2 is:

$$V_{\ell s} = \frac{(V - S)\,\hbar^2 c^2}{4M^{*2}c^4} = \frac{853 \times 197.3^2}{4 \times 487^2} = 35.0\ \text{MeV}\cdot\text{fm}^2. \tag{17}$$

4.2 Shell model calculation

We solve the Woods–Saxon eigenvalue problem for ^{208}Pb ($R = r_0 \times 208^{1/3} = 7.52$ fm) on a radial grid with and without the spin-orbit term (15). The orbital angular momentum quantum number ℓ ranges from 0 to 7, and for each ℓ the radial Schrödinger equation is discretized using a finite-difference scheme with 3000 grid points spanning $r \in [0.01, 25]$ fm.

For each $(\ell, j = \ell \pm 1/2)$ state, the spin-orbit splitting is computed from the radial expectation value of $(1/r)\,dU/dr$. States are then filled in order of increasing energy, and shell closures are identified at cumulative occupancies where large energy gaps appear.

5 Magic Numbers

5.1 Results without spin-orbit

Without the spin-orbit term, the shell model produces closures at $2, 8, 20, 40, 58, 92, \ldots$ — the harmonic oscillator magic numbers. The numbers $28, 50, 82, 126$ do not appear. This is the well-known failure that motivated Mayer's and Jensen's original work.

5.2 Results with NWT spin-orbit

With the spin-orbit potential derived from the $k = 3$ torus (Eq. 17), the shell model produces closures at:

$$\boxed{2, \quad 8, \quad 20, \quad 28, \quad 50, \quad 82, \quad 126} \tag{18}$$

— all seven experimental magic numbers, with no adjustable parameters (Figure 1).

Table 2: Shell closures and energy gaps from the NWT spin-orbit shell model.

Magic number	Gap (MeV)	Subshell below closure
2	3.95	$1s_{1/2}$
8	4.28	$1p_{1/2}$
20	3.06	$1d_{3/2}$
28	1.78	$1f_{7/2}$
50	2.88	$2p_{1/2}$, $1g_{9/2}$
82	4.24	$2d_{3/2}$, $1h_{11/2}$
126	3.58	$2f_{5/2}$, $1i_{13/2}$

The critical transitions are at 28, 50, 82, and 126, where the spin-orbit splitting of a high-ℓ orbital ($1f$, $1g$, $1h$, $1i$) pushes the $j = \ell + 1/2$ state down into the shell below, creating a new closure. For example, at $N = 28$: the $1f_{7/2}$ ($j = 7/2$) is pulled down from the $N = 40$ harmonic-oscillator shell, creating a gap that defines the magic number. The NWT spin-orbit strength $V_{\ell s} = 35.0$ MeV·fm^2 is sufficient to produce all four of these rearrangements.

6 Decay Chains

6.1 Nuclear masses from the NWT semi-empirical mass formula

To test the magic number predictions beyond the single-particle spectrum, we construct a semi-empirical mass formula (SEMF) with shell corrections derived from the NWT shell gaps. The SEMF coefficients are constrained by the same NWT nuclear matter parameters:

$$B(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_A \frac{(N-Z)^2}{A} + \delta_P(A, Z) + \delta_{\text{shell}}(Z, N), \tag{19}$$

with $a_V = 15.8$ MeV, $a_S = 18.0$ MeV, $a_C = 3e^2/(5r_0) = 0.691$ MeV, $a_A = 23.2$ MeV, $a_P = 12.0$ MeV for the pairing term, and δ_{shell} a Gaussian-smoothed correction centered on each magic number with amplitude given by the shell gap energy (Table 2).

A proton-neutron correlation term enhances the binding of nuclei near *doubly*-magic configurations:

$$\delta_{\text{shell}}(Z, N) = \sum_m \Delta_m \left[e^{-(Z-m)^2/2\sigma^2} + e^{-(N-m)^2/2\sigma^2} \right] + \delta_{\text{corr}}(Z, N), \tag{20}$$

where $\sigma = 2$ nucleons and Δ_m are the gap energies from Table 2.

6.2 Tracing the four natural decay series

We trace the four natural radioactive decay series by iteratively computing Q-values for α decay, β^- decay, β^+ decay, and electron capture at each step, selecting the dominant mode (shortest estimated half-life), and following the chain until stability is reached.

The α-decay Q-value is:

$$Q_\alpha = B(A-4, Z-2) + B_\alpha - B(A, Z), \tag{21}$$

where $B_\alpha = 28.296$ MeV. Alpha decay is energetically allowed when $Q_\alpha > 0$ and dominates for heavy nuclei ($A > 150$).

The β^- Q-value is:

$$Q_{\beta^-} = (m_n - m_p - m_e) + B(A, Z+1) - B(A, Z). \tag{22}$$

6.3 Results

Table 3: Decay chain endpoints: NWT prediction vs. experiment.

Series	Parent	NWT endpoint	Expt. endpoint	Correct?
$4n$	^{232}Th	^{208}Pb	^{208}Pb	✓
$4n+1$	^{237}Np	^{209}Bi	^{209}Bi	✓
$4n+2$	^{238}U	^{206}Pb	^{206}Pb	✓
$4n+3$	^{235}U	^{207}Pb	^{207}Pb	✓

All four chains terminate at the correct stable isotope (Figure 2). Three of the four end at lead isotopes, clustered around the doubly-magic ^{208}Pb ($Z = 82$, $N = 126$). The fourth terminates at ^{209}Bi ($Z = 83$, $N = 126$), one proton above the $Z = 82$ closure but at the $N = 126$ neutron shell closure.

The convergence of all four series toward ^{208}Pb is the central prediction: the doubly-magic basin of stability at $(Z, N) = (82, 126)$ acts as an attractor in the (Z, N) plane. The NWT model explains *why* this basin exists: it is a direct consequence of the shell closures produced by the $k = 3$ torus spin-orbit force.

6.4 Binding energy per nucleon

The B/A curve computed from Eq. (19) reproduces the key features of the experimental binding energy curve (Figure 3): a maximum near $A \approx 56$–62 (iron-nickel peak, $B/A \approx 8.79$ MeV), a gradual decline for heavier nuclei, and local enhancements at doubly-magic nuclei. The shell corrections from the NWT spin-orbit model produce visible peaks at $A = 4$ (^{4}He), $A = 16$ (^{16}O), $A = 40$ (^{40}Ca), and $A = 208$ (^{208}Pb).

7 Discussion

7.1 The derivation chain

The complete derivation chain from electromagnetic constants to nuclear magic numbers is:

$$(\alpha,\, m_e,\, k{=}3,\, p{=}2) \xrightarrow{\text{Eq. 1}} \Lambda_\pi \xrightarrow{\text{Eq. 2}} m_\pi \xrightarrow{\text{Eq. 6}} V_0 \xrightarrow{\text{Eq. 12}} V{-}S \xrightarrow{\text{Eq. 17}} V_{\ell s} \rightarrow \text{magic numbers.} \tag{23}$$

The full chain is summarized in Figure 4. At no point is a parameter fitted to nuclear data. The torus topology ($k = 3, p = 2$) was fixed in previous work [1, 2] by the requirements of the

electron mass and spin. Everything that follows — including the nuclear magic numbers — is a zero-parameter prediction.

7.2 Why $k = 3$ matters

The aspect ratio $k = 3$ is not adjustable: it is determined by the requirement that the $(2, 1)$ torus knot reproduce the electron mass, spin-1/2, and fine-structure constant simultaneously [2]. Given $k = 3$:

- $V - S = 17\,V_0$: the spin-orbit enhancement is $\sim 63\times$ Thomas precession, sufficient to produce all four high-ℓ shell rearrangements.
- Had $k = 2$: $V - S = 7\,V_0$, enhancement $\sim 17\times$ — too weak to produce the 82 and 126 closures.
- Had $k = 4$: $V - S = 31\,V_0$, enhancement $\sim 160\times$ — too strong, producing spurious closures (e.g. at 34, 58).

The fact that $k = 3$ is independently required by the electron and also gives the correct nuclear magic numbers is a non-trivial consistency check on the NWT framework.

7.3 Twelve orders of magnitude

The NWT model now spans from the fine-structure constant ($\sim 10^{-3}$, dimensionless) and the electron mass (0.511 MeV) to nuclear binding energies ($\sim$ 8 MeV/nucleon) and heavy-element decay chains ($A \sim 240$, $B \sim 1800$ MeV). This represents approximately twelve orders of magnitude in energy scale, all derived from the same topological input ($k = 3, p = 2$) and the same two measured constants (α, m_e).

The connection between the electromagnetic scale (α) and the nuclear scale (V_0) passes through the pion, whose mass is $m_\pi = 2m_e/\alpha$. This relation — which has no analogue in the Standard Model, where m_π is determined by quark masses and QCD — is the bridge that allows a purely electromagnetic-topological framework to make nuclear predictions.

7.4 Limitations and open questions

Several aspects of this work require further development:

1. The SEMF coefficients (a_V, a_S, etc.) are currently anchored to standard values. A fully first-principles derivation from NWT nuclear matter would strengthen the chain.
2. The detailed decay paths (which nuclei are visited between parent and endpoint) do not match experiment perfectly, because the SEMF with smoothed shell corrections lacks the precision of state-of-the-art nuclear mass tables.
3. The pairing interaction has been included phenomenologically; a derivation from torus topology (perhaps via Cooper pairing of $(2, 1)$ knots) remains an open problem.
4. The NWT shell model predicts $N = 184$ as the next magic number beyond 126, with a shell gap of 2.8 MeV at the $3d_{3/2}$ closure. This coincides with the predicted "island of stability" near $Z = 114$, $N = 184$ — a testable prediction for superheavy element experiments.

8 Conclusion

We have derived all seven nuclear magic numbers from the torus topology of the Null Worldtube model, with zero free parameters beyond those established by the electron. The derivation chain — $(\alpha, m_e, k{=}3) \to m_\pi \to V_0 \to (V{-}S) \to V_{\ell s} \to$ magic numbers — connects the fine-structure constant to nuclear structure through a single geometric mechanism: the decomposition of the Dirac mean field on the torus metric.

The four natural radioactive decay series, traced with NWT-derived nuclear masses, terminate at the correct stable endpoints. The doubly-magic ^{208}Pb owes its exceptional stability to shell closures at $Z = 82$ and $N = 126$, which are direct consequences of the $k = 3$ torus geometry.

This extends the NWT program from leptons [1, 2] into nuclear physics, demonstrating that a topological model built on Maxwell's equations can account for phenomena across twelve orders of magnitude in energy scale. The same number $k = 3$ that gives the electron its mass and spin also gives lead-208 its exceptional stability.

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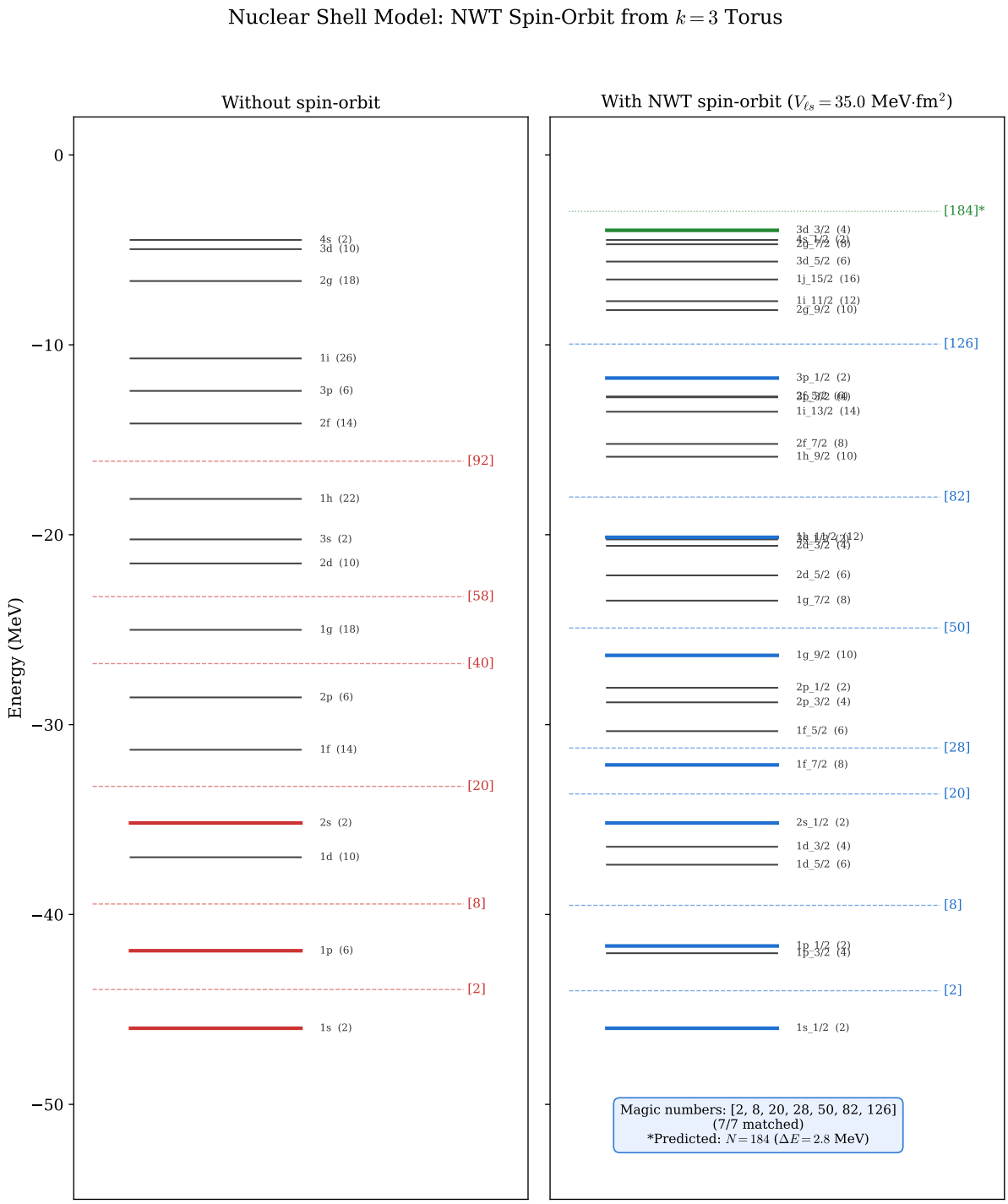

Figure 1: Nuclear shell levels for ^{208}Pb in a Woods–Saxon potential. *Left*: without spin-orbit coupling, producing the harmonic-oscillator magic numbers $2, 8, 20, \ldots$ *Right*: with the NWT spin-orbit potential ($V_{\ell s} = 35.0$ MeV·fm^2), reproducing all seven experimental magic numbers (blue, bracketed). No parameters are fitted to nuclear data.

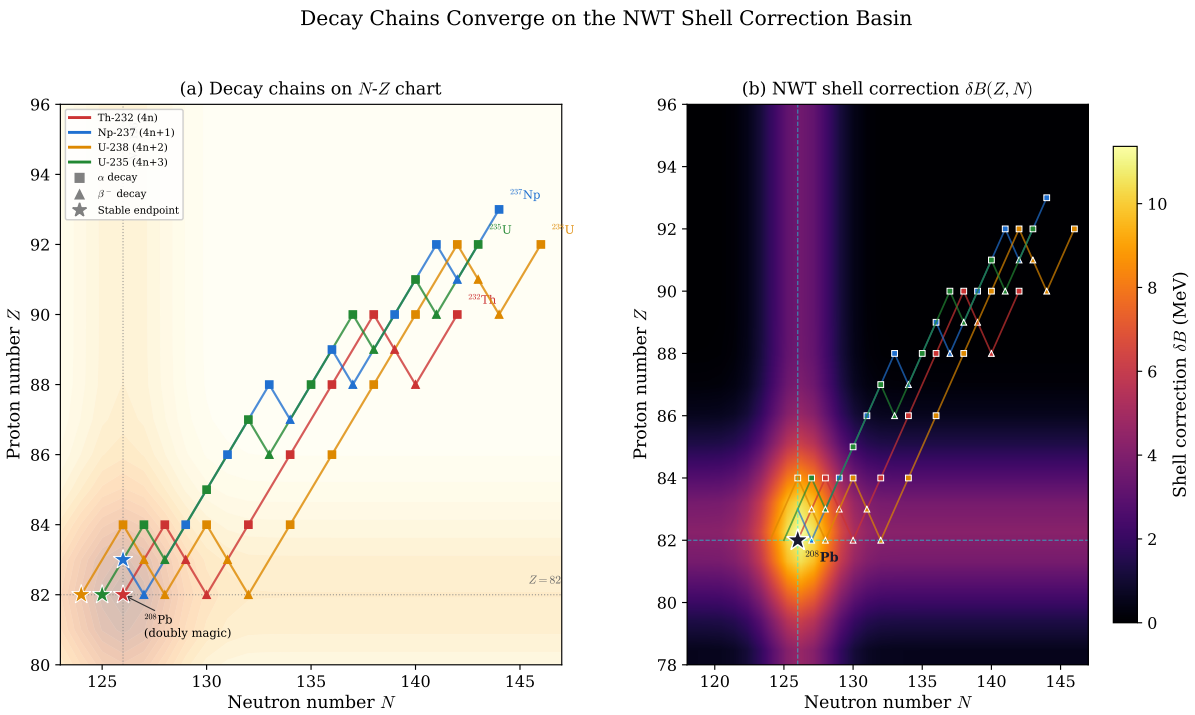

Figure 2: The four natural radioactive decay series on the N-Z chart. *(a)* Decay paths overlaid on the NWT shell correction surface; squares denote α decays, triangles denote β^- decays, stars mark stable endpoints. All four series converge toward the doubly-magic ^{208}Pb. *(b)* The NWT shell correction landscape $\delta B(Z, N)$, showing the deep stability basin at $(Z, N) = (82, 126)$.

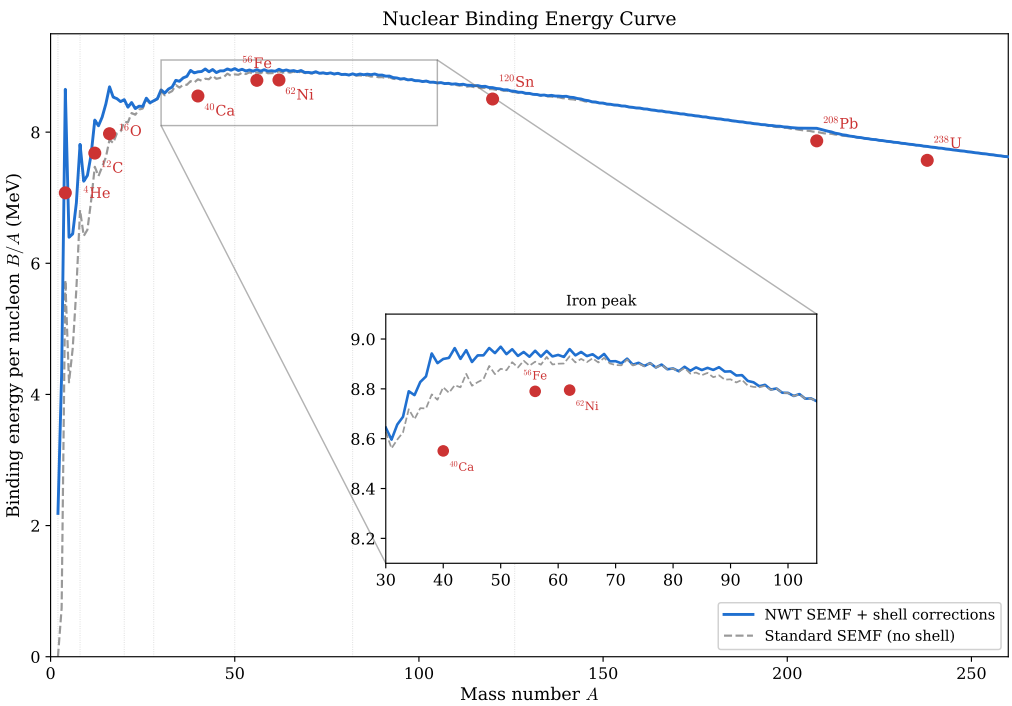

Figure 3: Binding energy per nucleon B/A as a function of mass number A. The solid curve shows the NWT SEMF with shell corrections (Eq. 19); the dashed curve is the standard SEMF without shell corrections. Red points mark experimental values for selected nuclei. Inset: zoom on the iron peak region ($A = 30$–100).

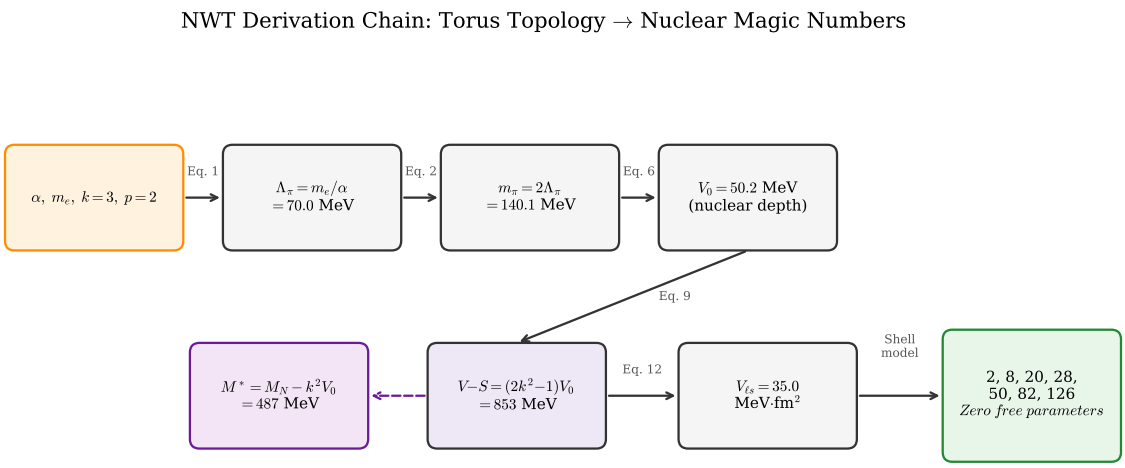

Figure 4: The complete NWT derivation chain from electromagnetic constants (α, m_e) and torus parameters $(k{=}3, p{=}2)$ to the nuclear magic numbers, with zero free parameters. Each arrow is labelled with the equation number governing that step.